Lam Fung Academy · LF Academy

Primary 5 • 5th Lesson • Student Handout

Special breakthrough in area trap

formulaComprehensive Review • SSPAHigh Frequencytrap • comprehensive questionstrategy • 65 minutes • 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5A, + Modern Education 5A, Units 1-4 (comprehensive review)

Core traps: • Inversion of operations during inverse calculation

SSPA correlation:

🔴 **High frequency** The area question in the sub-test is the hardest hit area, and we eliminate all common traps Prerequisite knowledge:

P4-P5 All area formulas (square/rectangular/parallelogram/triangle/trapezoid) The goal of this class:

❶ Comparative memory of three shape formulas ❷ Crack SSPA common traps one by one ❸ Establish a problem-solving strategy process ❹ Achieve zero mistakes in traps ❶ Comparative memory of three shape formulas ❷ Solve the common pitfalls of SSPA one by one ❸ Establish a problem-solving strategy process ❹ Achieve zero mistakes in traps

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions (total 5 questions, 5 minutes)—quickly determine mixed shapes

#	Question	Difficulty	Working Space
1	The parallelogram has a base of 9 cm and a height of 6 cm. Area = ?	🌱	
2	Triangular base 12 cm, height 5 cm. Area = ?	🌱	
3	The upper base of the trapezoid is 5 cm, the lower base is 9 cm, and the height is 4 cm. Area = ?	🌱	
4	The rectangle is 11 cm long and 6 cm wide. Its area = ? Perimeter = ? (Note: Area and perimeter must be written separately!)	🌿	
5	The side length of the square is 7 cm. Area = ? (Someone answered 28 cm²—what' s wrong?)	🌿	

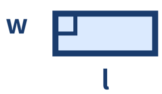
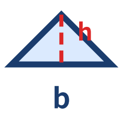
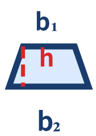
II.Core Knowledge + Worked Examples

Knowledge Point 1: General Review of Area Formula - Comparison of Three Shapes 🔴 SSPA Required Exam

① The following three shapes**When the base and height are the same |||SEP|||, the area relationship is fixed**② Memory formula: "Multiply the square square completely, the triangle needs half, the trapezoid has two bases and multiplies the height and then half"

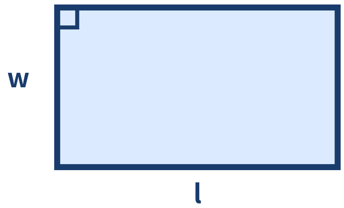
③ The unit must be written cm^2 / m^2, if you omit ^{2}, points will be deducted in the score test

③ The unit must be written in cm^2 / m^2. If ^{2} is omitted, points will be deducted in the test.

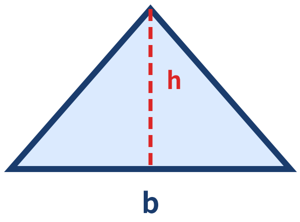
shape	icon	area formula	key pitfalls
parallelogram		**bottom × height**	Height is the vertical distance, not the hypotenuse
triangle		**bottom × height÷ 2**	It' s easiest to forget ÷2!
trapezoid		**(upper bottom + lower bottom) × height÷ 2**	Forgot ÷ 2. Using the wrong hypotenuse as the height, and inverting the upper and lower sides.

The area relationship of three shapes with the same base and height

parallelogram **triangle**

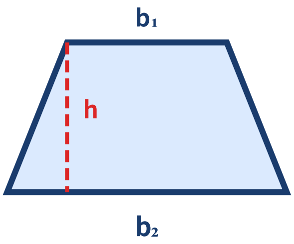


bottom x height = max



Bottom x height/2 = half

trapezoid



(a+b)xh/2

The base and height of all three are the same!

Example 1

The bases of parallelograms, triangles, and trapezoids are all 10 cm, and the heights are all 6 cm (the upper base of a trapezoid is 4 cm, and the lower base is 6 cm). Find the areas separately and compare the sizes.

Example 2

A parallelogram and a triangle have the same base and the same height. The area of a parallelogram is 48 cm². Area of triangle = ?

(When the base is the same and the height is the same, the area of the triangle is half of the parallelogram)

—— PAGE 3: KP1 continued + KP2 ——

Example 3

There is a trapezoid and a parallelogram whose heights are equal. The sum of the upper base + lower base of the trapezoid is equal to the base of the parallelogram. The area of a parallelogram is 60 cm². Trapezoidal area = ?

(upper base + lower base = base and height are equal → the area of the trapezoid is half of the parallelogram)

Knowledge point 2: Common SSPA area traps 🔴 High frequency points lost

1. **Unit conversion trap (T5) |||SEP|||: The question is given as m but the answer must be cm² → 1 m² = 10,000 cm² (not 100!)**Combined graphics trap (T4+T5) |||SEP|||: The size after division must be calculated from the original drawing, which may interfere with the length of the excess hypotenuse

2. **Reverse question type trap (T4) |||SEP|||: Find the base/height of the area, and the direction of operation is easy to reverse**÷2 forgetting trap (T5) |||SEP|||: The formulas for triangles and trapezoids must be ÷2, which is confused with parallelograms
3. **Perimeter area confusion (T5) |||SEP|||: The question asks about area, and students use the perimeter formula to answer**: Find the base/height of the area, and the direction of operation is easy to reverse.
4. **÷2 Forgotten Trap (T5)**: Triangle and trapezoid formulas must ÷2, confused with parallelogram
5. **Perimeter area confusion (T5)**: The question asked about area, and students answered using the perimeter formula.

T4 redundant information trap

Trapezoid: upper base 5, lower base 9, height 4

Hypotenuse length 6 →Superfluous!

The length of the hypotenuse is not used in the area formula

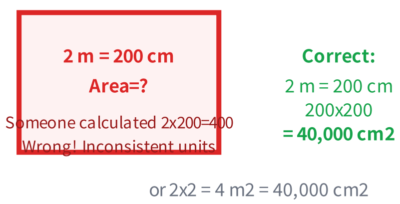
T5 graphic illusion trap

Area of triangle = base × height

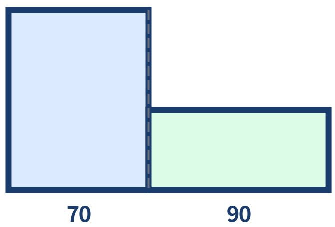
→ Leak ÷2!

The formula must be written completely: base × height ÷ 2

Trap comparison chart — unit conversion pit



Combined graphics – size calculation



Example 4

The upper base of the trapezoid is 5 cm, the lower base is 8 cm, the hypotenuse is 7 cm, and the height is 4 cm. Area = ?

(The question gives an extra 7cm for the hypotenuse - that is a T4 trap, no need to use it!)

Example 5

A rectangle is 2 m long and 150 cm wide. Area = ? cm²

(Unify the units first: 2 m = 200 cm → 200×150)

—— PAGE 4: KP3 + Layering 🌱 ——

Knowledge point 3: Area problem-solving strategy - four-step clearance method 🔴 SSPA

Step 1: Determine the shape— Square/rectangle/parallelogram/triangle/trapezoid/combined shape?
Step 2: Choose the right formula— Different shapes have different formulas, write them next to each other and then substitute
Step 3: Substitute carefully— Confirm that all units are consistent and eliminate redundant information
Step 4: Check the units— The area must be ² (cm², m²), and the answer is complete

Tips for solving the problem: "First determine the shape, second select the formula, substitute three into the fourth unit. Write the formula first and then do the algebra. The answer must be ²."

Example 6

Square perimeter 32 cm. Area = ?

(Trap! You cannot directly use 32 to calculate the area. First find the side length = 32÷4=8, and then calculate the area = 8×8=64 cm^2)

Comprehensive trap alert: The questions may mix T4 (superfluous information) + T5 (graphic illusion). For example, in the trapezoid question, the hypotenuse, perimeter, and units are mixed, which must be peeled off layer by layer!

III. Lesson Layered Synchronization Practice

Basic layer (required by everyone, total 5 questions)

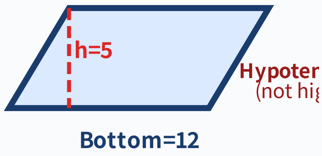
#	Question	Difficulty	Working Space
6	The parallelogram has a base of 10 cm and a height of 5 cm. Area = ?	🌱	
7	Triangular base 14 cm, height 6 cm. Area = ?	🌱	
8	The upper base of the trapezoid is 6 cm, the lower base is 12 cm, and the height is 5 cm. Area = ?	🌱	
9	The side length of the square is 9 cm. Write its**perimeter**and**area**respectively. Which one has²?	🌱	
10	The rectangle is 9 cm long and 5 cm wide. Area = ? Perimeter = ?	🌱	

Advanced layer (🧗🚀 choose do, total 5 questions)

#	Question	Difficulty	Working Space
11	Trapezoid: upper base 5 cm, lower base 9 cm, hypotenuse 8 cm, height 4 cm. Area = ? (Which data is not needed?)	🌿	
12	The area of the triangle is 36 cm^2 and the base is 9 cm. High = ? (reverse calculation: height = area × 2÷base)	🌿	
13	A rectangle is 3 m long and 2 m wide. Area = ? cm^2 (Trap: unit conversion! 3m=300cm, 2m=200cm)	🌿	
14	Square perimeter 36 cm. Its area = ? (Cannot directly replace 36! Find the side length first)	🌿	

15	L-shaped combination graphic: can be divided into two rectangles A (10×4 cm) and B (6×5 cm). Total area = ?	🌿	

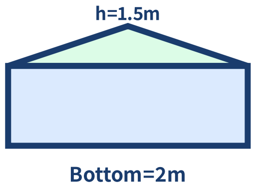
🌳 challenge layer (🚀 choose do, total 5 questions)

#	Question	Difficulty	Working Space
16	 The parallelogram has a base of 12 cm, an adjacent hypotenuse of 8 cm, and a height of 5 cm. Area = ? Perimeter = ? (For area, use base × height; for perimeter, use (base + hypotenuse) × 2—note that they use different sides!)	🌳	
17	The upper base of the trapezoid is 6 cm, the lower base is 10 cm, and the height is 5 cm. The base of a parallelogram is equal to the upper base + lower base of the trapezoid, and the height is the same as the trapezoid. How many times the area of a parallelogram is the area of a trapezoid? (parallelogram=16×5=80, trapezoid=(6+10)×5÷2=40 → 2 times)	🌳	
18	Triangular base 8 cm, height 6 cm. If the base remains unchanged and the height increases by 2 cm, how much more will the new area be than the original area? (original=24, new=8×8÷2=32, 8 cm^2 more)	🌳	
19	Trapezoidal area 72 cm^2, height 8 cm. The upper base is 1/2 of the lower base. How much are the upper and lower bases? (upper bottom + lower bottom =72×2÷8=18, set lower bottom=x upper bottom=0.5x → 1.5x=18)	🌳	
20	A parallelogram is bounded by 28 cm of iron wire (the length of each side is an integer cm and adjacent sides are not equal). What is the largest possible area? (Perimeter=28→Base+Hypotenuse=14, Area=Base×Height, combinations of different base heights need to be considered)	🌳	

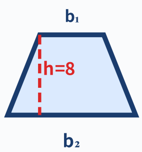
—— PAGE 6: Application questions ——

IV.areaapplicationquestion — SSPA comprehensivetrap training (total 8 questions)

#	Question	Difficulty	Working Space
21	The triangular garden bed has a base of 12 m and a height of 8 m. Area = ? m^2	🌱	
22	The terraced field is 15 m above ground, 25 m below ground, and 10 m high. Area = ? m^2	🌱	

23	A parallelogram wall has a base 6 m, adjacent sides 5 m, and a height 4 m. To paint, use 0.25 L per m^2. How many liters are needed in total? (5m is the hypotenuse—redundant information! Area=6×4=24 m^2)	🌿	
24	Square garden perimeter 48 m. If 5 flowers are planted per m^2, how many flowers can be planted in total? (First find the side length=48÷4=12, area=144 m^2)	🌿	
25	An L-shaped park is divided into: rectangle A (20 m × 15 m) and rectangle B (12 m × 8 m). Total park area = ? m^2	🌿	
26	The trapezoidal parking lot has an upper base of 30 m, a lower base of 50 m, and a height of 20 m. Each car occupies 25 m^2, how many can be parked at most?	🌳	
27	A triangle and a parallelogram have the same base and the same height. The area of the triangle is 25 m^2. Area of parallelogram = ?	🌳	
28	 A combined mural: a triangular section (base 2 m, height 1.5 m) beneath a rectangular section (2 m × 3 m). Total area = ? m^2	🌳	

V.⛰ Ultimate challenge area (total 3 questions, 🚀 choose do, SSPA finale + competition level)

#	Question	Difficulty	Working Space
⛰ 1	 Trapezoidal area 144 cm^2, height 8 cm. The upper base is 3 times larger than the lower base. How much are the upper and lower bases? (upper bottom + lower bottom = 144×2÷8=36, set lower bottom = x upper bottom = 3x → 4x=36 → x=9, upper = 27)	⛰	
⛰ 2	A polygon can be divided into: triangle (base 8 cm, height 5 cm) + rectangle (8 cm × 6 cm) + trapezoid (upper base 4 cm, lower base 8 cm, height 3 cm). Total area = ? **Calculate into three parts:** Triangle A = Base x Height/2 Rectangle B = Length Trapezoid C = (a+b)xh/2 總Area = A + B + C	⛰	

⛰ 3	After increasing the side length of the square by 3 cm, the new square has an area 57 cm^2 greater than the original square. The side length of the original square = ? (Suppose the original side length=x → $(x+3)^2-x^2=57$ → $x^2+6x+9-x^2=57$ → 6x=48 → x=8)	⛰	

—— PAGE 7: Homework + Common Mistakes ——

VI.Class afterhomework

Basic must-do (total 5 questions)

#	Question	Difficulty	Working Space
H1	The parallelogram has a base of 11 cm and a height of 6 cm. Area = ?	🌱	
H2	Triangular base 16 cm, height 5 cm. Area = ?	🌱	
H3	The upper base of the trapezoid is 7 cm, the lower base is 15 cm, and the height is 6 cm. Area = ?	🌱	
H4	Square perimeter 28 cm. Area = ? (Cannot directly replace 28! First find the side length = 28÷4=7)	🌿	
H5	The rectangle is 4 m long and 250 cm wide. Area = ? m^2 (250 cm = 2.5 m)	🌿	

For advanced, choose do (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	The trapezoid has an area of 84 cm^2, a height of 7 cm, and the lower base is twice as large as the upper base. How much are the upper and lower bases?	🌳	
H7	A triangle and a parallelogram have equal heights. The base of a triangle is 10 cm and the base of a parallelogram is 5 cm. How many times the area of a triangle is that of a parallelogram?	🌳	
H8	A rectangle whose length is twice its breadth. If perimeter is 36 cm, area = ?	⛰	

VII. The Lessoncorecommon errorsummary

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

- ☐ Identify all trap types in our hall
- ☐ Solve 🌱basic questions independently (100% correct)
- ☐ Challenge 🌿Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

#	common error	Correct Approach

1	**Triangle/Trapezoid Forget ÷2**(T5 Graphic Illusion)	Triangle = base × height**÷2**; trapezoid = (upper base + lower base) × height**÷2**
2	**Perimeter formula when area uses**(T5 confusion)	Area is "product" and perimeter is "sum". They are completely different.
3	**High use of wrong hypotenuse**(T5 graphic illusion)	Height =**vertical distance of the base**, not the length of the sloping side
4	**Unit conversion error**（T5）	1 m^2 = **10,000**cm^2 (not 100! Because 100×100)
5	**When calculating the area of the perimeter, directly substitute**(T4 redundant information)	First find the side length/bottom height from the perimeter → then find the area (must be done in two steps!)
6	**The direction of operation is reversed when searching inversely.**（T4）	Area×2÷base=height (triangle); area×2÷(upper base+lower base)=height (trapezoid)
7	**Area unit is missing²**	The answer for the area must be written in cm^2 / m^2. Points will be deducted if you fail to submit²!